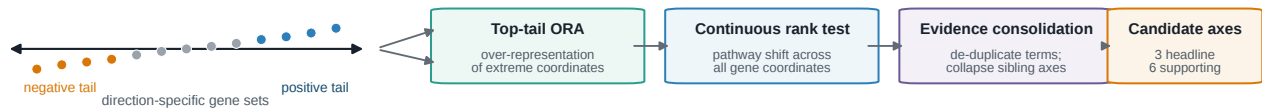


a Signed residual family coordinate

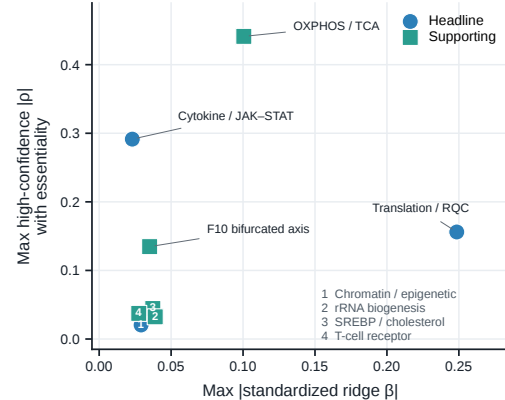


b Multiple evidence dimensions supporting curated axes

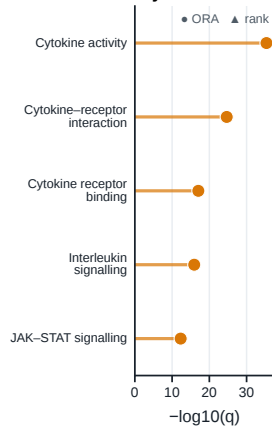
Cytokine / JAK-STAT	35.3	16.2	0.02	0.29	12.6
Chromatin / epigenetic	22.2	—	0.03	0.02	2.9
Translation / RQC	9.9	21.4	0.25	0.16	1.9
Chromatin-cytokine	23.8	21.8	0.04	0.13	2.9
OXPHOS / TCA	5.9	11.3	0.10	0.44	-3.5
Lysosome / ECM	3.1	4.4	0.04	0.13	2.9
rRNA biogenesis	4.0	3.4	0.04	0.03	1.0
SREBP / cholesterol	5.4	5.2	0.04	0.04	-2.3
T-cell receptor	3.7	3.7	0.03	0.04	-2.6
	ORA -log ₁₀ q	Rank -log ₁₀ q	Predictive β	Essentiality ρ	Universality index

Color scaled within each metric; cells show raw values.

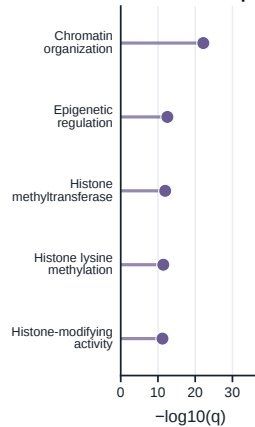
c Fitness relevance of parent families



d F14- Cytokine / JAK-STAT



e F3/F15 Chromatin / epigenetic



f F12 Translation / RQC

